

UNIVERSITY COLLEGE LONDON

MODULE CODE: COMP0078

MODULE NAME: COMP0078 – Supervised Learning

LEVEL: Undergraduate (Masters Level)

MOCK EXAMINATION

Duration: 3 Hours

Answer ALL FIVE (5) questions.

Marks for each part of each question are indicated in square brackets.

Standard calculators are permitted.

Question 1. (Weighted Ridge Regression and Kernelization)

Let $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n) \in \mathbb{R}^d \times \mathbb{R}$ be a training set. Let $X \in \mathbb{R}^{n \times d}$ denote the design matrix whose i -th row is $\mathbf{x}_i^\top$, and $\mathbf{y} = (y_1, \dots, y_n)^\top \in \mathbb{R}^n$ the output vector. Let $\lambda > 0$ be a regularization parameter.

Consider the *weighted* Tikhonov regularization problem with a positive definite diagonal weight matrix $\Gamma = \text{diag}(\gamma_1, \dots, \gamma_n)$ with $\gamma_j > 0$ for all j :

$$\min_{\mathbf{w} \in \mathbb{R}^d} \left\{ \frac{1}{n} \|X\mathbf{w} - \mathbf{y}\|^2 + \lambda \mathbf{w}^\top \Gamma \mathbf{w} \right\}. \quad (1)$$

- (a) Write out the gradient of the objective in (1) with respect to $\mathbf{w}$, set it to zero, and derive the closed-form solution $\mathbf{w}^*$. [3 marks]
- (b) Show that the solution $\mathbf{w}^*$ from part (a) can be written in the form

$$\mathbf{w}^* = \Gamma^{-1} X^\top \mathbf{c}$$

for some vector $\mathbf{c} \in \mathbb{R}^n$, and give an explicit expression for $\mathbf{c}$.

Hint: Rearrange the normal equations to isolate $\mathbf{w}^*$ and factor out $X^\top$. [4 marks]

- (c) Using the result from part (b), show that the prediction on a new test point $\mathbf{x} \in \mathbb{R}^d$ can be written as

$$\langle \mathbf{w}^*, \mathbf{x} \rangle = \sum_{i=1}^n c_i \tilde{k}(\mathbf{x}_i, \mathbf{x}),$$

where $\tilde{k} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is defined by $\tilde{k}(\mathbf{x}_i, \mathbf{x}) = \mathbf{x}_i^\top \Gamma^{-1} \mathbf{x}$. Identify what role $\tilde{k}$ plays. [4 marks]

- (d) Suppose instead we work in a feature space via a feature map $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^N$ and replace $\mathbf{x}_i$ with $\phi(\mathbf{x}_i)$. We also set $\Gamma = I_N$ (the identity). Derive the *dual* solution for ridge regression in feature space, expressing the prediction on a test point $\mathbf{x}$ purely in terms of a kernel function $K(\mathbf{x}_i, \mathbf{x}) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}) \rangle$. [5 marks]
- (e) Discuss when the dual formulation is computationally advantageous over the primal formulation, referencing the dimensions n (number of data points) and N (feature space dimension). What happens when ϕ corresponds to a Gaussian kernel? [4 marks]

TURN OVER

Question 2. (AdaBoost: Training Error Bound and Regularization)

Recall that AdaBoost maintains a distribution D_t over training examples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ with $y_i \in \{-1, +1\}$, and at each round $t = 1, \dots, T$: selects a weak learner $h_t : \mathbb{R}^d \rightarrow \{-1, +1\}$ minimizing the weighted error $\epsilon_t = \sum_{i=1}^m D_t(i) \mathbf{1}[h_t(\mathbf{x}_i) \neq y_i]$, assigns weight $\alpha_t = \frac{1}{2} \ln \frac{1-\epsilon_t}{\epsilon_t}$, and updates:

$$D_{t+1}(i) = \frac{D_t(i) \exp(-\alpha_t y_i h_t(\mathbf{x}_i))}{Z_t},$$

where Z_t is a normalization constant, starting from $D_1(i) = 1/m$.

- (a) Show by induction (or direct calculation) that after T rounds, the distribution satisfies

$$D_{T+1}(i) = \frac{\exp(-y_i \sum_{t=1}^T \alpha_t h_t(\mathbf{x}_i))}{m \prod_{t=1}^T Z_t}.$$

[3 marks]

- (b) Using the result from part (a) and the fact that $\mathbf{1}[y_i \neq \text{sign}(f(\mathbf{x}_i))] \leq \exp(-y_i f(\mathbf{x}_i))$ whenever $f(\mathbf{x}_i) = \sum_{t=1}^T \alpha_t h_t(\mathbf{x}_i)$, prove that the training error of the final classifier satisfies

$$\frac{1}{m} \sum_{i=1}^m \mathbf{1}[\text{sign}(f(\mathbf{x}_i)) \neq y_i] \leq \prod_{t=1}^T Z_t.$$

[3 marks]

- (c) Show that if each weak learner has weighted error $\epsilon_t \leq \frac{1}{2} - \gamma$ for some $\gamma > 0$, the normalization constant satisfies $Z_t = 2\sqrt{\epsilon_t(1-\epsilon_t)} \leq e^{-2\gamma^2}$, and conclude that the training error decays as $e^{-2\gamma^2 T}$.

Hint: Use the inequality $\sqrt{1-x} \leq e^{-x/2}$ for $x \in [0, 1]$.

[4 marks]

- (d) Discuss whether AdaBoost is susceptible to overfitting as $T \rightarrow \infty$. In your discussion, explain the role of the *margin* $y_i f(\mathbf{x}_i) / \sum_{t=1}^T \alpha_t$ and describe at least one practical strategy to regularize AdaBoost. Compare the structural difference between the ensemble produced by AdaBoost and that produced by bagging.

[5 marks]

TURN OVER

Question 3. (Stochastic Gradient Descent and Kernelization)

Let $\ell : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable convex loss function and let $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n) \in \mathbb{R}^d \times \mathbb{R}$ be a training set. Consider minimizing the regularized empirical risk:

$$F(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \ell(\mathbf{w}^\top \mathbf{x}_i, y_i) + \frac{\lambda}{2} \|\mathbf{w}\|^2,$$

where $\lambda > 0$.

- (a) Define what it means for a function $g : \mathbb{R}^d \rightarrow \mathbb{R}$ to be L -Lipschitz continuous. [2 marks]
- (b) **Stochastic Gradient Descent (SGD)** proceeds as follows: starting from $\mathbf{w}_1 = \mathbf{0}$, at each step $t = 1, 2, \dots, T$, sample an index $i_t \in \{1, \dots, n\}$ uniformly at random and update

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta_t \left(\ell'(\mathbf{w}_t^\top \mathbf{x}_{i_t}, y_{i_t}) \mathbf{x}_{i_t} + \lambda \mathbf{w}_t \right),$$

where ℓ' denotes the derivative of ℓ with respect to its first argument, and $\eta_t > 0$ is the step size.

Show that each iterate $\mathbf{w}_t$ can be written as a linear combination of the training inputs:

$$\mathbf{w}_t = \sum_{i=1}^n a_i^{(t)} \mathbf{x}_i$$

for some coefficients $a_i^{(t)} \in \mathbb{R}$. Proceed by induction on t , explicitly identifying the update rule for the coefficients $a_i^{(t)}$. [8 marks]

- (c) Using the representation from part (b), explain how the prediction $\mathbf{w}_t^\top \mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^d$ can be expressed purely in terms of inner products $\langle \mathbf{x}_i, \mathbf{x} \rangle$, and hence how one can “kernelize” SGD by replacing $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ with $K(\mathbf{x}_i, \mathbf{x}_j)$ for a reproducing kernel K . [4 marks]
- (d) Discuss the computational cost per iteration of kernelized SGD compared to standard SGD. Specifically, compare:
- the cost of updating the coefficients $a_i^{(t)}$,
 - the cost of computing the stochastic gradient in the original space versus in kernel form.

When is the kernelized form advantageous? [3 marks]

- (e) Let K be the Gaussian kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\beta \|\mathbf{x} - \mathbf{x}'\|^2)$. Explain why it is impossible to run SGD directly in the associated feature space $\phi(\mathbf{x})$, and why the kernelized SGD from part (c) resolves this issue. [3 marks]

CONTINUED

Question 4. (VC Dimension and Generalization via Uniform Convergence)

Let $\mathcal{X}$ be an input space, $\mathcal{Y} = \{-1, +1\}$, and H a hypothesis class of functions $h : \mathcal{X} \rightarrow \mathcal{Y}$. Let $\mathcal{D}$ be a distribution over $\mathcal{X} \times \mathcal{Y}$, and $S = ((x_1, y_1), \dots, (x_m, y_m))$ an i.i.d. sample from $\mathcal{D}^m$. Define the true risk $L_{\mathcal{D}}(h) = \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y]$ and the empirical risk $L_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}[h(x_i) \neq y_i]$.

- (a) Recall the definition of the **VC dimension** of H . [2 marks]
- (b) Consider the hypothesis class of *intervals* on the real line:

$$H_{\text{int}} = \{h_{a,b} : a \leq b \in \mathbb{R}\}, \quad h_{a,b}(x) = \begin{cases} +1 & \text{if } x \in [a, b], \\ -1 & \text{otherwise.} \end{cases}$$

Prove that $\text{VCdim}(H_{\text{int}}) = 2$.

Hint: First exhibit a set of size 2 that is shattered; then argue that no set of size 3 can be shattered. [5 marks]

- (c) State **Hoeffding's inequality** for a sum of m independent, bounded random variables $Z_1, \dots, Z_m \in [0, 1]$ with mean $\mu = \mathbb{E}[\sum_{i=1}^m Z_i]$. [2 marks]
- (d) Suppose H is a *finite* hypothesis class. Using a union bound and Hoeffding's inequality, prove that with probability at least $1 - \delta$ over the draw of S ,

$$\sup_{h \in H} |L_S(h) - L_{\mathcal{D}}(h)| \leq \sqrt{\frac{\ln(2|H|/\delta)}{2m}}.$$

[6 marks]

- (e) Using the result from part (d), show that the Empirical Risk Minimizer $h_S = \arg \min_{h \in H} L_S(h)$ satisfies

$$L_{\mathcal{D}}(h_S) \leq \min_{h \in H} L_{\mathcal{D}}(h) + 2\sqrt{\frac{\ln(2|H|/\delta)}{2m}}$$

with probability at least $1 - \delta$. [4 marks]

- (f) The bound in part (d) applies to finite classes. For infinite classes, explain informally how the **growth function** $\Pi_H(m) = \max_{C \subset \mathcal{X}, |C|=m} |H_C|$ and **Sauer's lemma** ($\Pi_H(m) \leq (em/d)^d$ when $d = \text{VCdim}(H) < \infty$) allow one to replace $|H|$ with an effective (polynomial) quantity, thereby extending the bound to infinite classes with finite VC dimension. Explain the bias-complexity tradeoff that arises when choosing H . [6 marks]

TURN OVER

Question 5. (Online Learning with the Hedge Algorithm)

Consider the online allocation setting: at each round $t = 1, \dots, T$, a learner selects a distribution $\mathbf{v}_t \in \Delta_n = \{(v_1, \dots, v_n) : v_i \geq 0, \sum_{i=1}^n v_i = 1\}$ over n experts, then nature reveals a loss vector $\ell_t \in [0, 1]^n$. The learner incurs expected loss $L_A(t) = \mathbf{v}_t \cdot \ell_t$.

The **Hedge algorithm** (with learning rate $\eta > 0$) initializes $v_{1,i} = 1/n$ for all i , and updates:

$$v_{t+1,i} = \frac{v_{t,i} e^{-\eta \ell_{t,i}}}{\sum_{j=1}^n v_{t,j} e^{-\eta \ell_{t,j}}}.$$

- (a) Define the **regret** of the learner after T rounds with respect to the best expert in hindsight, and explain why $O(\sqrt{T})$ regret is considered desirable for an online algorithm. [3 marks]
- (b) Let $\mathbf{u} \in \Delta_n$ be an arbitrary distribution over experts and define the KL divergence $d(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n u_i \ln(u_i/v_i)$. Show that

$$d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1}) = -\eta \mathbf{u} \cdot \ell_t - \ln \left(\sum_{i=1}^n v_{t,i} e^{-\eta \ell_{t,i}} \right).$$

[4 marks]

- (c) Using the inequalities $e^{-x} \leq 1 - x + x^2/2$ for $x \in \mathbb{R}$ and $\ln(1+x) \leq x$ for $x > -1$, show that

$$d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1}) \geq \eta(\mathbf{v}_t \cdot \ell_t - \mathbf{u} \cdot \ell_t) - \frac{\eta^2}{2}.$$

[5 marks]

- (d) By summing part (c) over $t = 1, \dots, T$ (telescoping) and using $d(\mathbf{u}, \mathbf{v}_1) \leq \ln n$ and $d(\mathbf{u}, \mathbf{v}_{T+1}) \geq 0$, prove the regret bound:

$$\sum_{t=1}^T \mathbf{v}_t \cdot \ell_t - \min_{i \in [n]} \sum_{t=1}^T \ell_{t,i} \leq \frac{\ln n}{\eta} + \frac{\eta T}{2}.$$

[3 marks]

- (e) Optimize the bound in part (d) over $\eta > 0$. State the optimal η^* and the resulting regret bound. Discuss what happens if the time horizon T is not known in advance. [5 marks]

END OF PAPER